



FINAL YEAR PROJECT • FINAL EVALUATION

An AI-Assisted Mental Health Support Platform

Compassionate, always-available support that bridges the gap between therapy sessions.

PROJECT GROUP • F25CS186

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The Problem We Are Solving

Mental health care is hard to access — and even harder to sustain between sessions.

Therapy is expensive and infrequent. Patients are left without support during the days and weeks between appointments.

Early warning signs are missed. Therapists lack visibility into a patient's mood and crisis moments outside the clinic.

Self-help is fragmented. Resources, mood logs, scheduling and records live in disconnected tools.



73%

of people cite cost & access as barriers to care

1 in 4

adults experience a mental-health condition

24/7

support gap between scheduled sessions

Objectives & Goals



Always-on AI companion

Give patients an empathetic, private space to talk and reflect between therapy sessions.



Safety first

Detect crisis and self-harm language and guide users toward professional help immediately.



Track mood & progress

Let patients log moods and goals, and give therapists clear insight into their journey.



Connect to real care

Verified therapists, appointment booking, secure chat and video — all in one platform.



Private & secure

Protect sensitive records with JWT auth, two-factor login and access-controlled data.

Meet Benzi — One Connected Platform



Patient App

AI companion, mood tracking, appointments, secure chat, self-help resources & goals.



Therapist Portal

Verification, client management, availability, reports and crisis alerts.



Admin Dashboard

Therapist approval, subscriptions, coupons, revenue and platform oversight.



Powered by the Benzi AI Engine

A locally-hosted, fine-tuned empathetic language model running on Ollama — paired with a crisis-detection safety layer and retrieval over the patient's own records. Private by design: the model runs on our own infrastructure.

System Architecture

A modular client-server architecture: React web apps talk to a Node.js / Express REST API and a Socket.IO real-time layer, backed by MongoDB and a Redis-powered job queue, with the Benzi AI engine served locally via Ollama.



Paste your System Architecture diagram here

Tip: export the architecture / component diagram and drop it into this frame

The Benzi AI Engine



Fine-tuned Empathetic Model

Base model

Qwen2.5 Instruct

Method

QLoRA fine-tuning (PEFT)

Served with

Ollama — runs locally

Training data

1,000 curated examples



Empathetic companion

Acknowledges feelings, reflects, and asks one gentle question — concise and supportive, never a diagnosis.



Crisis-detection safety layer

Flags self-harm and crisis language and steers users to their therapist or emergency help.



Retrieval (RAG) over records

Grounds replies in the patient's own notes and documents using vector embeddings.



Sentiment & mood analysis

Reads emotional tone to log mood trends and surface insights for therapists.

For Patients



Benzi AI Companion

Private, always-available empathetic chat with voice input support.



Mood Tracking

Daily mood logs that build a visual picture of emotional wellbeing.



Appointments

Browse verified therapists and book, reschedule or cancel sessions.



Secure Chat & Video

Real-time messaging and anonymous video meets with your therapist.



Self-Help Resources

Curated meditations, exercises and guidance for between sessions.



Goals & Progress

Set wellness goals and track achievements and progress over time.

For Therapists & Admins



Therapist Verification

Document upload and admin approval before therapists go live.



Client Management

Per-patient panels with history, notes and AI-generated context briefs.



Availability & Scheduling

Set working hours, services and pricing; sync with appointments.



Crisis Alerts

Instant email alerts when a patient shows crisis signals in chat.



Subscriptions & Billing

Stripe-powered plans, coupons and usage limits for therapists.



Revenue & Analytics

Admin dashboards for revenue, subscriptions and platform health.

Data Model & Data Flow



Paste your ER Diagram (ERD) here

Entities: User, Patient, Therapist, Appointment, Record, Message...



Paste your Data Flow Diagram (DFD) here

Level-1 DFD: auth, appointments, messaging, records & AI, billing

Technology Stack



Frontend

- React + Vite
- Tailwind CSS
- Socket.IO client
- React Router



Backend

- Node.js + Express
- MongoDB + Mongoose
- Redis + BullMQ
- Socket.IO



AI / ML

- Python · PyTorch
- Qwen2.5 + QLoRA
- Ollama (local)
- RAG embeddings



Integrations

- Stripe billing
- JWT + 2FA auth
- Nodemailer email
- Google Calendar

Application Screenshots



Paste Patient Dashboard & Mood

Patient web app



Paste Benzi AI Chat

AI companion conversation



Paste Therapist / Admin Dashboard

Management & analytics

Conclusion & Future Work



What we delivered

- A full-stack platform with patient, therapist and admin apps
- A locally-hosted, fine-tuned empathetic AI companion on Ollama
- A crisis-detection safety layer with real-time therapist alerts
- Mood tracking, appointments, secure chat, video & subscriptions
- Secure auth with JWT and two-factor login



Where we go next

- Mobile apps for iOS and Android
- Larger fine-tuning dataset & multilingual support
- Voice-first conversations with the AI companion
- Wearable & sleep data for richer mood insights
- Clinical validation studies with partner therapists



Thank You

Questions & Discussion

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